

Computer Organization vs. Architecture

Q. Define Computer Organization and Computer Architecture. Clearly distinguish between the two concepts using suitable examples.

> **Definition to Remember**

1. Computer Architecture (The "What")

- Deals with the attributes of a system visible to a programmer.

2. Computer Organization (The "How")

- Deals with the operational units and their interconnections.

3. Key Differences

Feature	Computer Architecture	Computer Organization

4. Real-World Example

The **Intel x86 family** architecture has remained largely compatible from the original 8086 up to modern processors (allowing old code to run). However, its **organization** (cache structures, pipelining, physical chips, clock speeds) has changed drastically over decades to improve performance.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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